		P × OD = 200 × OB sin 60° P = 170 N
5	B	[Method 1] distance travelled by A $= 4t - \frac{1}{2}(g\ sin\ sin\ 30°)t^2$ distance travelled by B $= \frac{1}{2}(g\ sin\ sin\ 30°)t^2$ total distance = $4t - \frac{1}{2}(g\ sin\ sin\ 30°)t^2 + \frac{1}{2}(g\ sin\ sin\ 30°)t^2 = 5$ therefore, t = 1.25 s [Method 2] speed of B relative to A = 0 – (–4) = 4 m s^{-1} time taken for B to travel 5 m = 5 / 4 = 1.25 m (or speed of A relative to B = 4 – 0 = 4 m s^{-1}, time taken for A to travel 5 m = 5 / 4 = 1.25 s)
6	D	If the constant time interval is T, distance travelled by each successive bead is $s = \frac{1}{2}g(nT)^2$